

Grade 7
Unit 4
Lesson 10 Practice

Name Answers
Date _____
Period _____

1. Tisa wants to cover the top of her circular button with a colorful fabric. The button has a diameter of 9 in . How much area needs to be covered by the fabric?
$$A = \pi r^2$$
$$= \pi 4.5^2$$
$$= 20.25\pi$$
$$= 63.617 \text{ in}^2$$
2. A farmer is installing a new fence for a circular pig pin. The diameter of the pin is 18.4 ft . How much fencing will be needed to enclose the pin?
$$C = \pi d$$
$$= \pi \cdot 18.4$$
$$= 57.805 \text{ ft}$$
3. A popular Ferris wheel is 300 feet tall. After one full trip around the Ferris wheel, how far will the person have travelled?
$$C = 300\pi$$
$$= 942.478 \text{ ft}$$
4. A circular rug measures $6\frac{3}{7} \text{ ft}$ from edge to edge of the rug. How much of the floor, in feet, is covered by the rug?
$$A = r^2\pi$$
$$= \left(3\frac{3}{14}\right)^2 \pi$$
$$= 10\frac{65}{196}\pi$$
$$= 32.458 \text{ ft}^2$$
$$\frac{1}{2}\left(6\frac{3}{7}\right) = r$$
$$3\frac{3}{14} = r$$
5. A radio tower broadcasts its signal at a range of 13.7 miles from the tower to its further point. How many square miles are covered by the range of this tower?
$$A = \pi r^2$$
$$= (13.7)^2\pi$$
$$= 187.69\pi$$
$$= 589.646 \text{ miles}^2$$

6. A construction crew has to put up a safety net around a large, circular opening in the ground that has a diameter of 31.1 ft. How much safety net is needed to ensure the opening is protected?

$$\begin{aligned}C &= \pi d \\&= 31.1\pi \\&= 97.704 \text{ ft}\end{aligned}$$

7. A helicopter blade is 54 ft in length. How far does the blade travel in one full rotation?

$$\begin{aligned}C &= \pi d \\&= 54\pi \\&= 169.646 \text{ ft}\end{aligned}$$

8. Gino has a clock with a minute hand that measures 9.4 inches. How far does the minute hand travel in *one* hour?

$$\begin{aligned}C &= \pi d & r &= 9.4 & d &= 2 \cdot 9.4 = 18.8 \\&= 18.8\pi \\&= 59.062 \text{ in}\end{aligned}$$

9. A circular round-a-bout is placed in a new neighborhood to regulate traffic. If the radius of the round-a-bout is 75 ft, how many feet of asphalt is needed to fill the round-about?

$$\begin{aligned}A &= \pi r^2 \\&= 75^2\pi \\&= 5625\pi \\&= 17,671.459 \text{ ft}^2\end{aligned}$$

10. In preparation for a snowstorm, Javier wants to wrap his tires with chains. If one tire has a diameter of 10.6 ft, what is the length, in feet, of chain needed to wrap the tire *one* time?

$$\begin{aligned}C &= \pi d \\&= 10.6\pi \\&= 33.301 \text{ ft}\end{aligned}$$